

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (EE) Examination

March – April 2018

EE242 Circuit Theory

Date: 28.03.2018, Wednesday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed.

a) Define the following terms. (Any seven).

[07]

1. Mesh
2. Transform Admittance
3. Node
4. Link
5. Loop
6. Tree
7. Transform Impedance
8. Twig
9. Branch

Q – 2 Draw the equivalent circuits across x-y terminals using Thevenin's & Norton's theorem [14]
of the diagram shown in **Fig 1**.

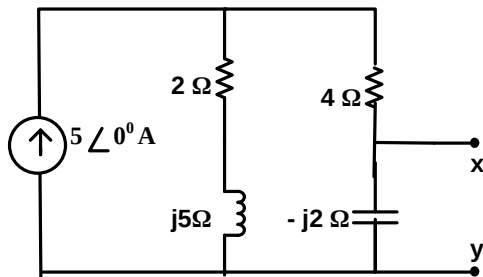


Fig.1 (Q-2)

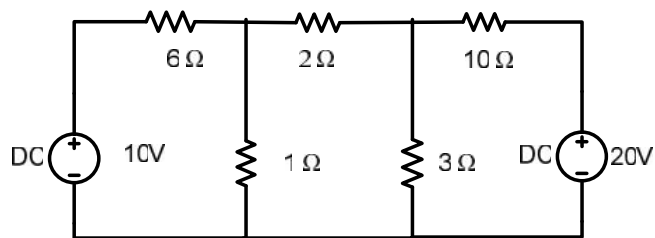


Fig.2 (Q-2(OR))

OR

Q – 2 Find all the branch current using Mesh Analysis and node voltage using Nodal Analysis method in the circuit shown in **Fig.2** using Cramer's rule. [14]

Q - 3 Attempt any two.

[14]

- a) Obtain the expression of $i(t)$ for step response of R- C Series circuit
- b) What are the different special signals are used in Circuit Theory? Define each input by giving its mathematical expressions and waveforms.
- c) The reduced incidence matrix of an oriented graph is given below.

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 \\ -1 & 0 & -1 & 0 & 0 \end{bmatrix}$$

- (i) Draw the graph.
- (ii) Prepare Incidence matrix
- (iii) How many trees are possible?

SECTION – II

Q - 5 Attempt any two.

[14]

- Classify the various types of special signals used in network analysis. Characterize each in detail by giving their mathematical expressions along with waveforms. Also derive their values in Time and Laplace domain.
- Represent the Y parameter in terms of (i) Z- parameter (ii) ABCD parameter.
- What will be the condition for symmetricity and reciprocity for Z parameter? Derive the condition by taking the two port network.

Q - 6 Attempt any two.

[14]

- What do you mean by initial conditions in the network? Why we need to find out the initial condition of the network? Write down all the steps to find out the initial condition parameters of the circuits.
- In the network shown in **Fig.3** capacitor C1 is charged to 100V in the polarity shown switch K is closed at time $t=0$. Determine i_1 , i_2 , $\frac{di_1}{dt}$, $\frac{di_2}{dt}$, $\frac{d^2i_1}{dt^2}$ & $\frac{d^2i_2}{dt^2}$ at $t=0_+$.
- (1) In a series R-L circuit, a unit step voltage $u(t-2)$ is applied the circuit parameters are $R=2\ \Omega$, $L=1\text{ H}$. At time $t=0$, find $i(t)$ assuming zero initial condition in the elements.
(2) In a series R-L-C circuit, a shifted doublet function is applied having $R=3\ \Omega$, $L=1\text{ H}$, $C=0.5\text{ F}$ at time $t=0$. Find $i(t)$ assuming zero initial condition in the elements.

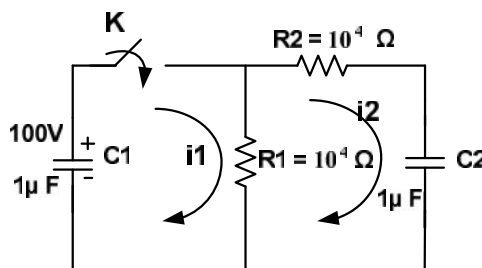


Fig. 3

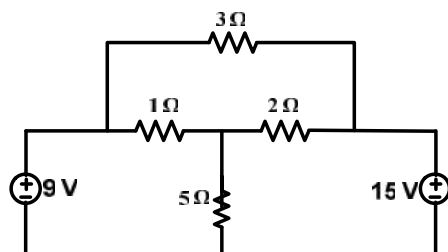


Fig. 4

Q - 7 Obtain ABCD parameters of the network shown in **Fig.4**.

07
